THE PROPORTION OF DIAGONALIZABLE MATRICES OVER A FINITE FIELD

SHEELA DEVADAS, PETER J. HAINE, & KEATON STUBIS

Abstract. We apply methods of combinatorics along with the orbit–stabilizer theorem to show that the proportion of diagonalizable $n \times n$ matrices over a finite field $\mathbf{F}_q$ approaches $1/n!$ as $q \longrightarrow \infty$.

0. Overview

Throughout this paper p denotes a prime, q a prime power, and n a positive integer. Over a finite field $\mathbf{F}_q$ there are q^{n^2} distinct $n \times n$ matrices. This paper completely answers the question of what proportion of these matrices are diagonalizable over $\mathbf{F}_q$ in the limit as $q \longrightarrow \infty$.

To wrap our heads around the problem, let us first consider the case of 2×2 diagonalizable matrices over $\mathbf{F}_p$.

Example A. The question we need to answer is *when is a* 2×2 *matrix over* $\mathbf{F}_p$ *diagonalizable?* If the eigenvalues are distinct and in $\mathbf{F}_p$, then there is a basis for $\mathbf{F}_p^2$ which consists of eigenvectors, so the matrix is diagonalizable. Therefore the matrices that are not diagonalizable have characteristic polynomials which are squares. If the characteristic polynomial is square, then the matrix is diagonalizable if and only if it is a scalar multiple of the identity (as the unique matrix conjugate to $\lambda \operatorname{id}_2$ is simply $\lambda \operatorname{id}_2$ itself, where id_2 denotes the 2×2 identity matrix and $\lambda \in \mathbf{F}_p$).

Thus we can count the matrices diagonalizable over $\mathbf{F}_p$ via the following method. We divide the matrices into two sets: multiples of the identity, and matrices with two distinct eigenvalues. There are p multiples of the identity. For the case of two distinct eigenvalues, we consider the number of ways to choose two linearly independent eigenvectors. We know that every nonzero vector in $\mathbf{F}_p^2$ is a multiple of one of

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \dots, \begin{pmatrix} 1 \\ p-1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Any pair of these vectors are linearly independent and, moreover, every pair of linearly independent vectors can be scaled to be a pair from this set. Hence, up to ordering, there are $p(p+1)/2$ choices of two linearly independent eigenvectors. Since there are $p(p-1)$ ways to choose two distinct eigenvalues, the number of diagonalizable matrices which are not multiples of the identity is $p^2(p-1)(p+1)/2$. The total number of matrices diagonalizable over $\mathbf{F}_p$ is then

$$\frac{p^2(p-1)(p+1)}{2} + p = \frac{p^4 - p^2 + 2p}{2}.$$

Since there are p^4 square 2×2 matrices over $\mathbf{F}_p$, as $p \longrightarrow \infty$ we see that the number of matrices over $\mathbf{F}_p$ which are diagonalizable over $\mathbf{F}_p$ approaches $1/2$.

Date: April 1, 2015.

The method described in Example A does not immediately generalize to high dimensional cases. To attack the problem in arbitrary dimension we take the ideas from Example A and extend them by applying more powerful tools from group theory and combinatorics. The well-known ***orbit–stabilizer theorem*** from abstract algebra (coupled with Lagrange's theorem) states that if G is a finite group and X a (nonempty) G-set, then for every $x \in X$,

$$\#G = \#O_x \cdot \#S_x,$$

where O_x denotes the orbit of x and S_x denotes the stabilizer of x [1, Pr. 6.9.2]. For a finite field $\mathbf{F}_q$, let $D_n(\mathbf{F}_q)$ denote the set of $n \times n$ matrices over $\mathbf{F}_q$ which are diagonalizable over $\mathbf{F}_q$. By definition, an $n \times n$ matrix is diagonalizable just in case it is conjugate to a diagonal matrix, that is, it is in the orbit of a diagonal matrix under the action of $\mathrm{GL}(n, \mathbf{F}_q)$ on the matrix ring $\mathrm{Mat}_n(\mathbf{F}_q)$ by conjugation. Write $\mathcal{O}$ for the set of orbits in $\mathrm{Mat}_n(\mathbf{F}_q)$ which contain a diagonal matrix. For any diagonal matrix D, write $O_D \in \mathcal{O}$ for the orbit containing D, and S_D for the stabilizer of D. Then by the orbit–stabilizer theorem

$$(\star) \qquad \#D_n(\mathbf{F}_q) = \sum_{O_D \in \mathcal{O}} \#O_D = \#\mathrm{GL}(n, \mathbf{F}_q) \sum_{O_D \in \mathcal{O}} \frac{1}{\#S_D}.$$

Hence to compute $\#D_n(\mathbf{F}_q)$, it suffices to compute the orders of the orbits containing a diagonal matrix (or, equivalently, the stabilizer of a representative from each orbit). Example B below illustrates how to apply this method for the low-dimensional case of $n = 3$ over the field $\mathbf{F}_2$.

Example B. Consider $\mathrm{Mat}_3(\mathbf{F}_2)$. It is easy to show that there are $\binom{4}{3} = 4$ distinct orbits under the action of $\mathrm{GL}(3, \mathbf{F}_2)$ which contains a diagonal matrix. Two of these orbits are the orbit of the zero matrix and the orbit of the identity matrix, each of which only contains one element. The other two orbits are the orbits of the diagonal matrices D with one eigenvalue equal to 1 and a repeated eigenvalue of 0, and D' with one eigenvalue equal to 0 and a repeated eigenvalue of 1. By simple matrix multiplication it is easy to see that the there is a bijection between matrices in the stabilizer of D and elements of $\mathrm{GL}(2, \mathbf{F}_2)$. Lemma 1.1 of § 1, which computes $\#\mathrm{GL}(n, \mathbf{F}_q)$ for any positive integer n and any finite field $\mathbf{F}_q$, shows that $\#\mathrm{GL}(2, \mathbf{F}_2) = 6$, hence the stabilizer of D contains 6 elements. The same is true for D'. Lemma 1.1 also shows that $\#\mathrm{GL}(3, \mathbf{F}_2) = 168$, hence using $(\star)$ we see that

$$\#D_3(\mathbf{F}_2) = \#\mathrm{GL}(3, \mathbf{F}_2)\left(\frac{1}{\#\mathrm{GL}(3, \mathbf{F}_2)} + \frac{1}{\#\mathrm{GL}(3, \mathbf{F}_2)} + \frac{1}{6} + \frac{1}{6}\right) = 58.$$

In § 1 we show that the size of an orbit containing a diagonal matrix can be written as a rational polynomial in q, hence applying $(\star)$ shows that $\#D_n(\mathbf{F}_q)$ can be written as a polynomial in q. This reduces the problem to describing the limiting behavior of a rational function. In § 2 we apply these technologies to prove the main result of this paper.

Theorem (The Main Result). *As* $q \longrightarrow \infty$,

$$\frac{\#D_n(\mathbf{F}_q)}{\#\mathrm{Mat}_n(\mathbf{F}_q)} \longrightarrow \frac{1}{n!}.$$

Acknowledgments. We thank Clark Barwick for his direction, guidance, and useful conversations.

1. Elementary Combinatorics & the Size of an Orbit

In this section we compute the number of diagonalizable matrices over a finite field. We use this computation, along with a combinatorial gadget called a *composition* to give a convenient formula for the total number of $n \times n$ diagonalizable matrices over $\mathbf{F}_q$ as a polynomial in q.

As a preliminary piece of the puzzle, in order to study the action of $\mathrm{GL}(n, \mathbf{F}_q)$ on $\mathrm{Mat}_n(\mathbf{F}_q)$, we need to determine the size of $\mathrm{GL}(n, \mathbf{F}_q)$. This follows from a simple combinatorial argument.

1.1. **Lemma.** *For any finite field $\mathbf{F}_q$ and any positive integer n, we have*

$$\#\mathrm{GL}(n, \mathbf{F}_q) = \prod_{i=0}^{n-1}(q^n - q^i).$$

Proof. We use the fact that a matrix is invertible if and only if its determinant is nonzero. Given an $n \times n$ matrix, for its determinant to be nonzero the only restriction on the first column is that it is nonzero, so there are $q^n - 1$ choices for this column. The only restriction on the second column is that it is not a multiple of the first, so there are $q^n - q$ choices for the second column. In general, the only restriction on the i^{th} column is that it is not a linear combination of the first $i-1$ columns, so there are $q^n - q^{i-1}$ choices for the i^{th} column. Hence in total there are $\prod_{i=0}^{n-1}(q^n - q^i)$ matrices over $\mathbf{F}_q$ with nonzero determinant. □

Since we are concerned with enumerating the orbits which contain a diagonal matrix, we need to know when two diagonal matrices lie in the same orbit — the following lemma tells us precisely when this happens.

1.2. **Lemma.** *Suppose k any field. A pair of $n \times n$ diagonal matrices over k are conjugate just in case they have the same eigenvalues counting multiplicity.*

Proof. The forward implication is obvious. For the converse statement we may assume, without loss of generality, that none of the eigenvalues of A and B are repeated. Let $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n$ denote the diagonal entries of A and B, respectively (ordered so that α_m and β_m are the m^{th} diagonal entries of A and B, respectively). Define a matrix P in the following way: if $\alpha_i = \beta_j$, then $p_{ji} = 1$, and otherwise $p_{ji} = 0$. Then, by construction, P is a permutation matrix, hence invertible over any field. Moreover it is clear by construction that $PA = BP$, so A and B are conjugate. □

1.3. **Proposition.** *Suppose D an $n \times n$ diagonal matrix over $\mathbf{F}_q$ with (distinct) eigenvalues $\lambda_1, \dots, \lambda_m$, with respective multiplicities $c_1, \dots, c_m$. The size of the orbit O_D of D under the action of $\mathrm{GL}(n, \mathbf{F}_q)$ on $\mathrm{Mat}_n(\mathbf{F}_q)$ by conjugation is*

$$\#O_D = \frac{\prod_{i=0}^{n-1}(q^n - q^i)}{\prod_{j=0}^{m}\prod_{i=0}^{c_j-1}(q^{c_j} - q^i)}.$$

Proof of Proposition 1.3. Write S_D for the stabilizer of D. The action of $\mathrm{GL}(n, \mathbf{F}_q)$ on $\mathrm{Mat}_n(\mathbf{F}_q)$ by conjugation corresponds to a change of basis. Sublemma 1.3.1 below shows that a matrix fixes D under conjugation if and only if it preserves all of the eigenspaces of D. Write V_j for the eigenspace of D corresponding to the eigenvalue λ_j. Then $\mathbf{F}_q^n$ decomposes as the direct sum $V_1 \oplus \cdots \oplus V_m$, and for each j we have $V_j \cong \mathbf{F}_q^{c_j}$. The number of new bases preserving all of the eigenspaces is the product of the numbers of bases for each eigenspace. Note that the number of bases for $\mathbf{F}_q^{c_j}$ is $\#\mathrm{GL}(c_j, \mathbf{F}_q)$.[†] Thus by Lemma 1.1 the size of S_D is

$$\#S_D = \prod_{j=1}^{m} \prod_{i=0}^{c_j-1} (q^{c_j} - q^i).$$

Hence, by orbit–stabilizer theorem,

$$\#O_D = \frac{\#\mathrm{GL}(n, \mathbf{F}_q)}{\#S_D} = \frac{\prod_{i=0}^{n-1}(q^n - q^i)}{\prod_{j=1}^{m} \prod_{i=0}^{c_j-1} (q^{c_j} - q^i)}. \qquad \square$$

1.3.1. **Sublemma.** *Suppose k a field and $D \in \mathrm{Mat}_n(k)$ a diagonal matrix. An invertible matrix P preserves all of the eigenspaces of D just in case P and D commute.*[‡]

Proof of Sublemma 1.3.1. First suppose P an invertible matrix that commutes with D. Suppose λ an eigenvalue of D with corresponding eigenspace V, and let $v \in V$. Then we see that

$$DPv = PDv = P(\lambda v) = \lambda Pv,$$

which shows that $Pv \in V$.

On the other hand, suppose P an invertible matrix that preserves all of the eigenspaces of D. Enumerate the eigenspaces of D as $V_1, \ldots, V_m$, and write λ_j for the eigenvalue of D corresponding to V_j. Then since D is diagonal, k^n decomposes as the direct sum $V_1 \oplus \cdots \oplus V_m$. Since P preserves all of the eigenspaces of D, for any $1 \leq j \leq m$ and any $v \in V_j$, we have $Pv \in V_j$. Hence $DPv = \lambda_j Pv$. On the other hand, we also see that

$$PDv = P(\lambda_j v) = \lambda_j Pv,$$

therefore $DPv = PDv$. Since $k^n = V_1 \oplus \cdots \oplus V_m$, we see that $DPu = PDu$ for all $u \in k^n$, which shows that P and D commute. $\square$

The following amazing fact is crucial to our argument — as we will see shortly, it reduces the problem of computing the limit

$$\lim_{q\to\infty} \frac{\#D_n(\mathbf{F}_q)}{\#\mathrm{Mat}_n(\mathbf{F}_q)}$$

to computing the limit as $q \longrightarrow \infty$ of a rational function of q.

1.4. **Lemma.** *With the same hypotheses as Proposition 1.3, the size of the orbit of an $n \times n$ diagonal matrix D over $\mathbf{F}_q$ is a rational polynomial in q.*

First we need a little technical sublemma as well as a bit of notation.

1.4.1. **Sublemma.** *Suppose $P = f/g$, for some $f, g \in \mathbf{Z}[x]$. If $P(\ell)$ is an integer for infinitely many integers $\ell \in \mathbf{Z}$, then P is in $\mathbf{Q}[x]$.*

Proof of Sublemma 1.4.1. Use polynomial long division to write

$$f(x) = \alpha(x)g(x) + r(x),$$

for some polynomials $\alpha \in \mathbf{Q}[x]$ and $r \in \mathbf{Z}[x]$, such that $\deg r < \deg g$. If α is not identically zero, let γ denote the product of the denominators of all of the nonzero coefficients of α,

[†]Here we do not need to account for scaling or ordering as those operations yield distinct elements of the stabilizer.

[‡]The astute reader will notice that this is not the most general result possible and that its proof only uses a subset of the hypotheses listed. However, this is the result that we care about, and a more general result is of little consequence.

otherwise, set $\gamma = 1$. Then $\gamma\alpha \in \mathbf{Z}[x]$. For each integer ℓ such that $P(\ell) \in \mathbf{Z}$, we see that that $\gamma r(\ell)/g(\ell)$ is an integer. Thus the hypotheses of the sublemma tell us that there are infinitely many $\ell \in \mathbf{Z}$ such that $\gamma r(\ell)/g(\ell)$ is an integer. However, since $\deg r < \deg g$ we see that

$$\lim_{x\to\pm\infty} \frac{r(x)}{g(x)} = 0,$$

which shows that r is identically zero. Hence $P = \alpha$, which shows that $P \in \mathbf{Q}[x]$. □

1.5. **Notation.** Suppose k a field and $A_1, \dots, A_m$ square matrices over k, where each A_j is in $\mathrm{Mat}_{c_j}(k)$, and write $n = \sum_{j=1}^m c_j$. We write $A_1 \oplus \cdots \oplus A_m$ for the ***matrix direct sum***, that is, the $n \times n$ the block diagonal matrix

$$A_1 \oplus \cdots \oplus A_m \coloneqq \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix},$$

where zero entries have been omitted for notational simplicity.

Proof of Lemma 1.4. First notice that for any field k and any finite sequence of positive integers $c_1, \dots, c_m$ there is an embedding

$$\prod_{j=1}^m \mathrm{GL}(c_j, k) \hookrightarrow \mathrm{GL}(\textstyle\sum_{j=1}^m c_j, k)$$

given by the assignment

$$(A_1, \dots, A_m) \rightsquigarrow A_1 \oplus \cdots \oplus A_m.$$

In the case that $k = \mathbf{F}_q$ and $\sum_{j=1}^m c_j = n$, all of the general linear groups are finite, so the size of $\prod_{j=1}^m \mathrm{GL}(c_j, \mathbf{F}_q)$ divides the size of $\mathrm{GL}(n, \mathbf{F}_q)$. By Lemma 1.1, the size of $\prod_{j=1}^m \mathrm{GL}(c_j, \mathbf{F}_q)$ is simply

$$\prod_{j=1}^m \#\mathrm{GL}(c_j, \mathbf{F}_q) = \prod_{j=1}^m \prod_{i=0}^{c_j-1} (q^{c_j} - q^i),$$

so applying Sublemma 1.4.1 proves the result. □

For our main argument we need a combinatorial tool that totally captures the idea of decomposing a diagonal matrix into diagonal blocks corresponding to its eigenvalues. This tool is called a *composition*. Informally a composition of a positive integer n is a way of writing n as a sum of positive integers, where the order in which we write the terms in the sum matters.

1.6. **Definition.** Suppose m and n positive integers. A ***composition of n of length m*** is an m-tuple c of positive integers, written as

$$c = [c_1, \dots, c_m],$$

such that $\sum_{j=1}^m c_j = n$. Write $\mathrm{Comp}(n)$ for the set of compositions of n, and $\mathrm{Comp}_m(n)$ for the set of compositions of n of length m.

1.7. **Remark.** We can think of a composition of n of length m is a multiset of $\{1, \dots, n\}$ of size m, where the order of the terms actually matters. Another way to think of a composition is as a partition in which order does matter.

Note that $\mathrm{Comp}_m(n)$ is empty for $m > n$.

In order to enumerate the elements of $D_n(\mathbf{F}_q)$, we want to put diagonal matrices into a canonical block from with the eigenvalues "ascending" down the diagonal and look at the orbits corresponding to these canonical diagonal matrices — in an ordered field such as $\mathbf{R}$ we can do this by using the order relation on the field. Since a finite field is not ordered, we cannot use this approach, which makes it harder to count diagonal matrices in a consistent manner. To get around this issue we make the following convention.

1.8. **Convention.** For each prime power q, we fix an arbitrary total ordering to the underlying set U_q of $\mathbf{F}_q$.

Note that the particular order relation actually does not matter at all — it is simply a combinatorial tool to aid in our counting. We use this ordering in the following way to look distinguish special diagonal matrices in a *canonical form* given by the total ordering on U_q.

1.9. **Definition.** Suppose O an orbit in $\mathrm{Mat}_n(\mathbf{F}_q)$ which contains a diagonal matrix, where the eigenvalues of the matrices in O are $\lambda_1 < \cdots < \lambda_m$ (ordered as elements of U_q) with respective multiplicities $c_1, \ldots, c_m$. We call the diagonal block matrix

$$\lambda_1 \,\mathrm{id}_{c_1} \oplus \cdots \oplus \lambda_m \,\mathrm{id}_{c_m},$$

where id_{c_j} denotes the $c_j \times c_j$ identity matrix, the ***canonical matrix representing*** O.

1.10. **Theorem.** *The number of $n \times n$ matrices with entries in $\mathbf{F}_q$ which are diagonalizable over $\mathbf{F}_q$ is*

$$\#D_n(\mathbf{F}_q) = \sum_{m=1}^{n} \sum_{c \in \mathrm{Comp}_m(n)} \binom{q}{m} \frac{\prod_{i=0}^{n-1}(q^n - q^i)}{\prod_{j=1}^{m} \prod_{i=0}^{c_j-1}(q^{c_j} - q^i)}.$$

Proof. Let $C_n(\mathbf{F}_q) \subset D_n(\mathbf{F}_q)$ denote the subset of canonical matrices. Notice that $\#C_n(\mathbf{F}_q)$ is the number of orbits containing a diagonal matrix. Define a map $\phi \colon C_n(\mathbf{F}_q) \longrightarrow \mathrm{Comp}(n)$ by the assignment

$$\lambda_1 \,\mathrm{id}_{c_1} \oplus \cdots \oplus \lambda_m \,\mathrm{id}_{c_m} \rightsquigarrow [c_1, \ldots, c_m].$$

Notice that the fiber of a composition $[c_1, \ldots, c_m]$ consists of all of the canonical diagonal matrices of the form $\lambda_1 \,\mathrm{id}_{c_1} \oplus \cdots \oplus \lambda_m \,\mathrm{id}_{c_m}$, for all possible choices of $\lambda_1 < \cdots < \lambda_m$. Since there are $\binom{q}{m}$ ways to choose m distinct eigenvalues from a set of q elements, we see that

$$\#\phi^{-1}([c_1, \ldots, c_m]) = \binom{q}{m}.$$

By Proposition 1.3, if A and B are any two canonical matrices such that $\phi(A) = \phi(B)$, then the orbits of A and B have the same size. Hence Proposition 1.3 tells us that the total number of matrices lying in an orbit which contains a diagonal matrix whose canonical representative maps to a given composition $[c_1, \ldots, c_m]$ under ϕ is

$$\#\phi^{-1}([c_1, \ldots, c_m]) \cdot \#O_D = \binom{q}{m} \frac{\prod_{i=0}^{n-1}(q^n - q^i)}{\prod_{j=1}^{m} \prod_{i=0}^{c_j-1}(q^{c_j} - q^i)},$$

where O_D is the orbit of any canonical matrix D such that $\phi(D) = [c_1, \ldots, c_m]$. Summing over to all possible compositions gives the formula for $\#D_n(\mathbf{F}_q)$. □

Example 1.11 below illustrates the formula given in Theorem 1.10 in action.

1.11. **Example.** Let us go back to Example B of §0 and once again consider $\mathrm{Mat}_3(\mathbf{F}_2)$. Since $2 < 3$, the binomial coefficient $\binom{2}{3}$ is zero, so Theorem 1.10 shows that there are only two types of compositions to consider — the length-one composition $[3]$ which corresponds to the diagonal matrices and the two length-two compositions $[1,2]$ and $[2,1]$, which correspond to the matrices D and D' of Example B. The composition $[3]$ yields $\binom{2}{1} = 2$ diagonalizable matrices, and each of the compositions $[1,2]$ and $[2,1]$ yield

$$\binom{2}{2}\frac{(2^3-1)(2^3-2)(2^3-2^2)}{(2^1-1)(2^2-1)(2^2-2)} = 28$$

diagonalizable matrices. Hence we see that

$$\#D_3(\mathbf{F}_2) = 1 + 1 + 28 + 28 = 58.$$

This agrees with the brute-force computation outlined in Example B.

2. The Proportion of Diagonalizable Matrices

Lemma 1.4 tells us that the orbit O_D of an $n\times n$ diagonal matrix D over $\mathbf{F}_q$ is a polynomial in q. This, coupled with Theorem 1.10 and the fact that the binomial coefficients

$$\binom{q}{m} = \frac{1}{m!}q(q-1)\cdots(q-(m-1)),$$

for $0 \le m \le q$, are polynomials in q tells us that $\#D_n(\mathbf{F}_q)$ is simply a polynomial in q. A slightly subtle but not totally irrelevant technical point (partially illustrated in Example 1.11) is that the polynomial given by $\#D_n(\mathbf{F}_q)$ is varies for $q < n$, but stabilizes for $q \ge n$. The reason for this is the fact that $\binom{q}{m} = 0$ for $m > q$. When $q \ge n$, none of the binomial coefficients vanish, so the same polynomial in q gives the formula for $\#D_n(\mathbf{F}_q)$ in this range. Since we are only concerned with the limiting behavior of $\#D_n(\mathbf{F}_q)$, this does not affect our computation of the limit

$$\lim_{q\to\infty}\frac{\#D_n(\mathbf{F}_q)}{\#\mathrm{Mat}_n(\mathbf{F}_q)},$$

but, as a result, the statements of the following theorems are slightly more technical than one would hope. Using the fact that $\#\mathrm{Mat}_n(\mathbf{F}_q) = q^{n^2}$ reduces our problem to describing the limiting behavior of a rational function.

2.1. **Notation.** We write $f_n(q) \coloneqq \#D_n(\mathbf{F}_q)$ when we wish to regard the number of diagonalizable matrices over $\mathbf{F}_q$ as a polynomial in q, as described in the preceding paragraph.

2.2. **Theorem.** *Let q be any prime power. For each positive integer $n \le q$, the leading term of $f_n(q)$ is $q^{n^2}/n!$.*

Proof. It is clear that the degree of the term of f_n corresponding to a composition $[c_1,\dots,c_m]$ is

$$\deg\binom{q}{m} + \deg\#O_D = n^2 + m - \sum_{j=1}^{m} c_j^2,$$

where O_D is the orbit of any canonical diagonal matrix D which maps to $[c_1,\dots,c_m]$ under the map ϕ (from the proof of Theorem 1.10). Since the c_j are all positive integers, we see that

$$\sum_{j=1}^{m} c_j^2 \ge m.$$

Therefore the degree of f_n is at most n^2. Moreover, this inequality is sharp — the degree n^2 term of f_n corresponds to the unique length-n composition $[1,\dots,1]$. Thus, the leading term of f_n is comes from

$$\binom{q}{n}\prod_{i=0}^{n-1}\frac{q^n-q^i}{(q-1)^n}.$$

For $0 \le i \le n-1$ we can factor $q^n - q^i$ as

$$q^n - q^i = q^i(q^{n-i} - 1) = q^i(q-1)\sum_{\ell=0}^{n-i-1} q^\ell$$

Therefore

$$\binom{q}{n}\prod_{i=0}^{n-1}\frac{q^n - q^i}{(q-1)^n} = \frac{q(q-1)\dots(q-n+1)}{n!}\prod_{i=0}^{n-1} q^i \sum_{\ell=0}^{n-i-1} q^\ell. \tag{2.2.1}$$

From this simplification it is easy to see that the coefficient of q^{n^2} in (2.2.1) is $1/n!$, hence the leading term of f_n is $q^{n^2}/n!$. ☐

The main result of the paper now follows as a simple corollary.

2.2.1. **Corollary** (The Main Result). *As $q \longrightarrow \infty$,*

$$\frac{\#D_n(\mathbf{F}_q)}{\#\mathrm{Mat}_n(\mathbf{F}_q)} \longrightarrow \frac{1}{n!}.$$

Proof. By Theorem 1.10, we see that

$$\frac{\#D_n(\mathbf{F}_q)}{\#\mathrm{Mat}_n(\mathbf{F}_q)} = \frac{f_n(q)}{q^{n^2}}.$$

By Theorem 2.2, when $q \ge n$, the leading term of $f_n(q)$ is $q^{n^2}/n!$, so in this range

$$f_n(q) = q^{n^2}/n! + g(q),$$

for some polynomial g of degree strictly less than n^2. Taking the limit as $q \longrightarrow \infty$ we see that

$$\lim_{q\to\infty}\frac{\#D_n(\mathbf{F}_q)}{\#\mathrm{Mat}_n(\mathbf{F}_q)} = \lim_{q\to\infty}\frac{q^{n^2}/n! + g(q)}{q^{n^2}} = \frac{1}{n!}.$$ ☐

Many algebraists today find it useful to contemplate the non-existent "field with one element". Setting $q = 1$ in the polynomial f_n ends up producing an interesting result which has a meaningful interpretation under this viewpoint — if there were a field with only one element, in any dimension the unique square matrix over that field would be diagonalizable.

2.3. **Proposition.** *For each positive integer n, the polynomial f_n has the property that $f_n(0) = 0$ and $f_n(1) = 1$.*

Proof. For each $m \ge 1$ we know that $\binom{0}{m} = 0$. Then since each term of the polynomial $f_n(q)$ is the product of the binomial coefficient $\binom{q}{m}$ with a polynomial in q, we see that $f_n(0) = 0$.

Similarly, for $m \ge 2$, the binomial coefficient $\binom{1}{m}$ is zero. Therefore the only terms in $f_n(1)$ which are nonzero are the terms corresponding to $m = 1$, that is, the compositions of length one. The only composition of n of length one is $[n]$. The canonical matrices mapping to $[n]$ under ϕ are simply the scalar multiples of the identity. For all q, there are q such matrices, each contained in an orbit of size 1. Therefore the term in $f_n(q)$ corresponding to the composition $[n]$ is equal to q as a polynomial. Thus we see that $f_n(1) = 1$. ☐

References

[1] Michael Artin, *Algebra*, 2nd ed., Pearson, 2010.

Massachusetts Institute of Technology, Department of Mathematics, 77 Massachusetts Avenue, Cambridge, MA 02139-4307, USA

E-mail address, corresponding author: phaine@mit.edu